

DR.BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,LONERE

Supplementary Winter Examination– 2023

Course:- TY.B.Tech Branch: Mechanical Engineering Sem.:- VI

Subject Name & Code: - Manufacturing Processes II (BTMC601_Y22)

Max. Marks: 60 Date:-16-01-2024 Duration:-3Hr.

Instructions to the Students

1. All the questions are compulsory.
2. Each question carries 12 marks.
3. Illustrate your answers with neat sketches, diagram, etc., wherever necessary.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks (6×2=12)
Solve any Two of the following:		
a) What are the key components of grinding wheels, and how do factors like bond types and wheel grade influence the grinding process?	Analyse	6
b) Explain the significance of grinding-wheel wear, grinding ratio, and the role of dressing, truing, and shaping in grinding operations.	Evaluate	6
c) How does the grindability of materials impact wheel selection in abrasive machining? Provide examples of materials with specific grinding challenges.	Apply	6
Solve any Two of the following:		
a) Describe the geometry of single-point cutting tools and discuss how it affects the metal cutting process.	Understand	6
b) What are the forces acting on a cutting tool during metal cutting and how is specific cutting energy related to plowing force and the "size effect"?	Apply	6
c) Compare the theories of Ernst and Merchant with Lee and Shaffer concerning chip thickness in metal cutting.	Analyse	6
Solve any Two of the following:		
a) Analyze the role of temperature in metal cutting, including heat generation, temperature distribution, and the effect of cutting speed on temperatures.	Analyse	6
b) Explain the different forms of tool wear in metal cutting and the criteria used for determining tool life.	Apply	6
c) Briefly discuss major classes of cutting tool materials and their basic requirements. What role does cutting fluid play in machining processes?	Understand	6

Solve any Two of the following:

(6×2=12)

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| a) Provide an overview of methods used in the production of metal powders and explain how particle size, shape, and distribution impact the final product. | Understand | 6 |
| b) Describe the compaction process of metal powders, focusing on isostatic pressing and the significance of sintering. | Apply | 6 |
| c) What are the secondary and finishing operations involved in processing powder metals, and how do they contribute to the final product's properties? | Evaluate | 6 |

Solve any Two of the following:

(6×2=12)

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| a) Explore various extrusion processes in the production of plastics, including miscellaneous techniques and polymer reinforcing fibers. | Understand | 6 |
| b) Briefly explain injection molding, blow molding, and rotational molding in the context of processing plastics. | Apply | 6 |
| c) Highlight advantages or unique characteristics that distinguish Solid-phase Forming of plastics from other forming processes. | Apply | 6 |